OS & SP

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SKILL – 19

1.Skill19.c

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GNU nano 8.7.1 skill19.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

pid_t foreground_pid = -1;

/*
 * SIGINT handler
 * Ctrl+C sends SIGINT to the shell.
 * The shell forwards it to the foreground child.
 */
void handle_sigint(int sig)
{
    (void)sig;

    if (foreground_pid > 0) {
        printf("\nShell: Forwarding SIGINT to foreground process\n");
        kill(foreground_pid, SIGINT);
    } else {
        printf("\nShell: Ignoring SIGINT\n");
    }
}

/* SIGTERM handler */
void handle_sigterm(int sig)
{
    (void)sig;
    printf("\nShell received SIGTERM\n");
}

int main(void)
{
    char input[256];

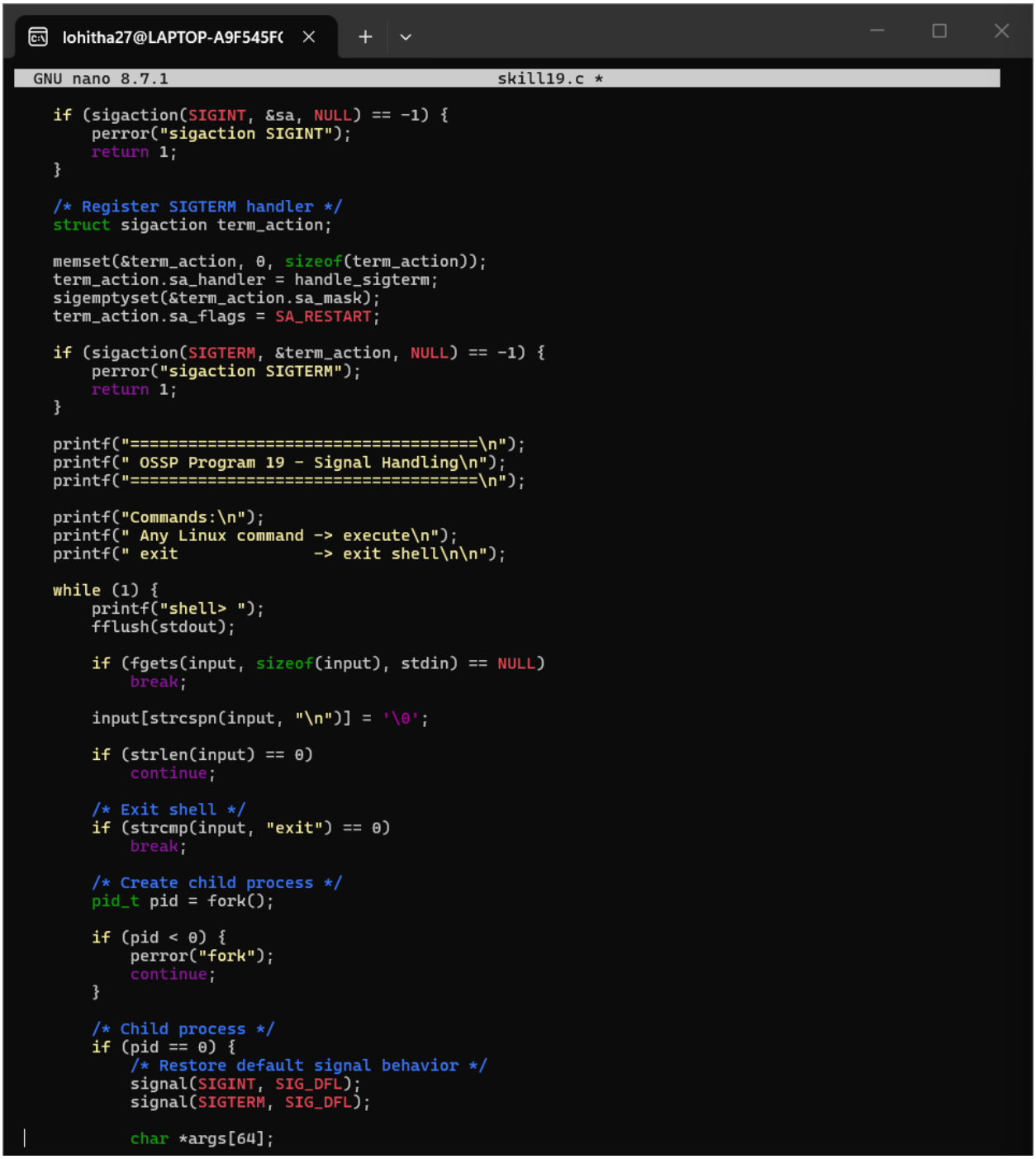
    /* Register SIGINT handler */
    struct sigaction sa;

    memset(&sa, 0, sizeof(sa));
    sa.sa_handler = handle_sigint;
    sigemptyset(&sa.sa_mask);
    sa.sa_flags = SA_RESTART;

    if (sigaction(SIGINT, &sa, NULL) == -1) {
        perror("sigaction SIGINT");
        return 1;
    }

    /* Register SIGTERM handler */
    struct sigaction term_action;

    memset(&term_action, 0, sizeof(term_action));
    term_action.sa_handler = handle_sigterm;
    sigemptyset(&term_action.sa_mask);
    term_action.sa_flags = SA_RESTART;
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memset(&term_action, 0, sizeof(term_action));
term_action.sa_handler = handle_sigterm;
sigemptyset(&term_action.sa_mask);
term_action.sa_flags = SA_RESTART;
if (sigaction(SIGTERM, &term_action, NULL) == -1) {
perror("sigaction SIGTERM");
return 1;
}
printf("==================================\n");
printf(" OSSP Program 19 - Signal Handling\n");
printf("==================================\n");
printf("Commands:\n");
printf(" Any Linux command -> execute\n");
printf(" exit -> exit shell\n\n");
while (1) {
printf("shell> ");
fflush(stdout);
if (fgets(input, sizeof(input), stdin) == NULL)
break;
input[strcspn(input, "\n")] = '\0';
if (strlen(input) == 0)
continue;
/* Exit shell */
if (strcmp(input, "exit") == 0)
break;
/* Create child process */
pid_t pid = fork();
if (pid < 0) {
perror("fork");
continue;
}
/* Child process */
if (pid == 0) {
/* Restore default signal behavior */
signal(SIGINT, SIG_DFL);
signal(SIGTERM, SIG_DFL);
char *args[64];

```
            char *args[64];
            int argc = 0;

            char *token = strtok(input, " \t");

            while (token != NULL && argc < 63) {
                args[argc++] = token;
                token = strtok(NULL, " \t");
            }

            args[argc] = NULL;

            if (argc == 0)
                _exit(0);

            /* Execute command */
            execvp(args[0], args);

            perror("execvp");
            _exit(1);
        }

        /* Parent process */
        else {
            foreground_pid = pid;
            int status;
            if (waitpid(pid, &status, 0) == -1) {
                perror("waitpid");
            } else {
                foreground_pid = -1;
                if (WIFEXITED(status)) {
                    printf("Process exited with status %d\n",
                           WEXITSTATUS(status));
                }
                else if (WIFSIGNALED(status)) {
                    printf("Process terminated by signal %d\n",
                           WTERMSIG(status));
                }
            }
        }
    }
    printf("\nShell terminated.\n");
    return 0;
}
```

OUTPUT:

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill19.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill19.c -o skill19
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill19
====================================
 OSSP Program 19 - Signal Handling
====================================
Commands:
 Any Linux command -> execute
 exit              -> exit shell

shell> pwd
/home/lohitha27/SKILL
Process exited with status 0
shell> exit

Shell terminated.
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```